

OPINION



Cross-reactivity in hypersensitivity reactions to contrast agents: new classification and guide for clinical practice

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Recent studies focusing on hypersensitivity reactions (HSR) after iodinated contrast media (ICM) administration have shown overall incidence rates of 0.15 to 0.69%, with severe reactions occurring in 0.005 to 0.013% of cases [1]. For gadolinium-based contrast agents (GBCA), HSR rates are lower (0.15–0.40%), with severe reaction rates of 0.002–0.004% in the general population and 0.033% in a population undergoing cardiac MRI [2]. Patients with a history of an HSR to CM may experience a recurrence upon re-administration [3–5]. In cases where a mild HSR is limited to the skin, the risk of developing a moderate to severe HSR after a new application of ICM is less than 1% [6]. Recurrence rates are higher in groups with more severe previous HSR, especially among patients whose initial reaction was anaphylactic [7]. In addition, if a reaction occurs again, the risk of it being severe is higher [4]. Therefore, in patients with a history of a prior severe reaction, re-administration of ICM is considered a relative contraindication.

Recommendations for patients with a history of HSR to contrast media (CM) have evolved in recent years. If possible, an attempt should be made to perform a study without CM or to use an alternative imaging modality, provided that the quality of the examination is not compromised. The effectiveness of premedication as a preventive measure has been questioned and is now deemed ineffective according to various European guidelines [8, 9]. In many instances, breakthrough reactions can occur despite the use of premedication [10, 11], with an incidence ranging from 1.2% to 46% [6, 12, 13]. It is noteworthy that breakthrough reactions can develop both after re-administration of the same culprit ICM

responsible for the previous reaction and when using another alternative ICM empirically. There is growing evidence supporting the greater safety of recommending alternative CM [14, 15]. However, a significant challenge in selecting safe alternatives lies in their high and variable cross-reactivity (CR), both between ICM and GBCA. Despite few recent studies assessing such CR [3, 16–20], results remain inconclusive, and a recognizable CR pattern has not been established.

The best outcomes are achieved when the choice is guided by a prior allergy assessment using skin tests, supplemented by provocation tests with the CM when necessary [8, 21–24]. It is not uncommon to lack an allergic study prior to a new administration of CM. This could be due to a shortage of specialists in drug allergy or inadequate facilities for provocation testing. Additionally, even if tests are accessible, delays in conducting them can occur. The allergic study ideally should be performed within a period of two to six months from the original reaction [25–27]. The effectiveness of these studies decreases if conducted too soon, as the patient may be in a refractory phase. However, when the delay exceeds this period, the efficacy of the study diminishes further. This is not only due to the loss of sensitization over time [22, 28] but also because of the reduced reliability of the patient's data, including failures in the description of symptom type and severity, as well as identification of the implicated CM [29].

The choice of an alternative CM without the support of an allergic study is complex, as cross-reactivity between them is high and variable among individuals, without being able to identify a reliable safety pattern. In fact, there are no randomized controlled studies comparing cross-reactivity, and the data previously reported are based on predominantly retrospective observational studies. To facilitate an empirical selection, some authors have attempted to classify CM into homogeneous groups

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Table 1 Classifications of non-ionic iodine-based contrast media according to sidechains**Lerondeau et al (2016) [31]**

- Group A: iohexol, iomeprol, ioversol, iodixanol, and iopamidol, sharing 2 identical N-(2,3-dihydroxypropyl) carbamoyl sidechains.
- Group B: iobitridol.

Schrijvers et al (2018) [26]

- Group A: iohexol, iomeprol, ioversol, iodixanol, and iopromide, all sharing a N-(2,3-dihydroxypropyl) carbamoyl sidechain.
- Group B: iobitridol and iopamidol.

Srisuwatchari et al (2022) [33]

- Group A: iohexol, iomeprol, ioversol, iodixanol, iopromide and iopamidol with at least 1 N-(2,3-dihydroxypropyl) carbamoyl group.
- Group B: iobitridol.

Seo et al (2023) [32]

- Iohexol, iomeprol, ioversol, iodixanol and iopromide, share a benzyl sidechain with an N-(2,3-dihydroxypropyl) carbamoyl group.
- Iopromide and iobitridol have an N-(2,3-dihydroxypropyl)-N-methyl-carbamoyl group.
- Iopamidol does not possess either a N-(2,3-dihydroxypropyl) carbamoyl group or N-(2,3-dihydroxypropyl)-N-methyl-carbamoyl group.

based on molecular similarity. Regarding ICM, for patients with a previous immediate HSR of any severity, using an alternative ICM without a common carbamoyl side chain reduced recurrent HSR during subsequent exposures [30]. It has been hypothesized that the ability to induce an HSR could be related to the presence or absence of a similar carbamoyl sidechain, although this theory lacks substantial data support. Nevertheless, despite all classifications being based on that same assumption [26, 31–33], the composition of the groups is different, and sometimes contradictory (Table 1). In an attempt to integrate the diverse proposals, we suggest a classification that enhances intragroup structural homogeneity, considering the 2D composition of each molecule (Table 2).

Group A would include iohexol, iomeprol, ioversol, and iodixanol, all sharing an N-(2,3-dihydroxypropyl) carbamoyl sidechain (classic carbamoyl sidechain). Group B would comprise iobitridol, with a N-(2,3-dihydroxypropyl)-N-methyl-carbamoyl sidechain (modified carbamoyl sidechain). Group C would include iopromide, exhibiting mixed characteristics, having both a classic and a modified carbamoyl sidechain. Finally, Group D consists of iopamidol, that has a different formulation featuring two 1-N,3-N-bis(1,3-dihydroxypropan-2-yl) sidechains (Fig. 1).

When choosing an alternative ICM empirically without access to allergic study results, we propose the following approach, considering that CR could be fundamentally based on chemical structure:

In the case of a known ICM:

When the implicated ICM is from Group A: Prioritize the use of one from Group B or D.

When the implicated ICM is from Group B: Prioritize the use of one from Group A or D.

When the implicated ICM is from Group C: Prioritize the use of Group D.

When the implicated ICM is from Group D: Prioritize the use of one from Group A, B, or C.

In the case of an unknown ICM:

There would be a higher probability that the implicated CM would be from Group A (greater number of ICM): Prioritize the use of one from Group B or D. The use of Group C is not recommended due to high CR with Group A.

Unfortunately, previous studies have noted that the tolerance pattern does not fit with this chemical structure, likely because in allergic HSR, the epitope may extend beyond the carbamoyl chain in 2D, and the presence of conformational epitopes in the molecule's 3D structure must be considered [7]. Additionally, the influence of molecular structure may have less significance in activating innate immunity mechanisms responsible for non-allergic HSR.

Different studies have suggested that Group D (iopamidol) can be an alternative, as it has a relatively favourable safety profile, although less favourable than expected based on its greater chemical difference [8]. In Group B (iobitridol), methylation of its carbamoyl chain creates a slight difference compared to other ICM. However, it has been found to have a high tolerance in patients with previous HSR [34, 35], with a confirmed different sensitization pattern in *in vitro* studies [36]. Therefore, considering both the chemical differences and results from clinical studies, we think, like other authors, that iobitridol may be a good choice when selecting an alternative ICM empirically [37–39].

Limited data exists regarding GBCA, primarily because they have been utilized for a shorter period compared to ICM, and furthermore, they induce fewer HSR [40]. However, it appears appropriate to include current information, along with some basic recommendations concerning an empirical selection of GBCA in the event of

Table 2 Classifications of nonionic iodine-based contrast media and gadolinium-based contrast agents

NON-IONIC IODINE-BASED CONTRAST MEDIA			
Group A	Group B	Group C	Group D
N-(2,3-dihydroxypropyl)-carbamoyl sidechains	N-(2,3-dihydroxypropyl)-N-methyl-carbamoyl sidechains	N-(2,3-dihydroxypropyl)-carbamoyl AND N-(2,3-dihydroxypropyl)-N-methyl-carbamoyl sidechains	1-N,3-N-bis(1,3-dihydroxypropan-2-yl) sidechains
Iohexol	Iobitridol	Iopromide	Iopamidol
Iodixanol			
Iomeprol			
Ioversol			

GADOLINIUM-BASED CONTRAST AGENTS		
Group A	Group B	Group C
DOTA (tetra-azacyclododecane tetra-acetic acid)	Variants of DO3A (Sodium 2,2',2''-(1,4,7,10-tetraazacyclododecane-1,4,7-triyl) triacetate)	Pyclen (3,6,9,15-tetrazabicyclo[9.3.1]pentadeca-1(15),11,13-trien
Gadoterate meglumine	Gadobutrol (with BT-DO3A) Gadoteridol (with HP-DO3A)	Gadopidlenol

a previous HSR. A recurrence rate of reaction up to 12% has been reported when a different GBCA with the same molecular structure was administered, and a 6% rate when a GBCA with a different molecular structure was used [41]. Therefore, theoretically, the use of a linear GBCA could be prioritized in the case of an HSR with a macrocyclic one, and vice versa. However, GBCAs used in the EU are almost exclusively limited to macrocyclic agents due to restrictions imposed by the European Medicines Agency on the use of linear agents because of the risk of nephrogenic systemic fibrosis and gadolinium deposition. Thus, in this article, we focus on grouping by chemical similarity only in the macrocyclic agents. Currently, there are four macrocyclic GBCAs available: one ionic (gadoterate meglumine) and 3 non-ionic (gadobutrol, gadoteridol, and gadopidlenol). They all have in common the presence of a chelating agent whose function is to prevent gadolinium toxicity in vivo.

If the ability to induce an HSR could reside within the chelating agent, we could classify macrocyclic GBCAs into three groups (Table 2):

- Group A: The chelating agent is DOTA, including gadoterate meglumine.
- Group B: The chelating agents are variants of DO3A, including gadobutrol (with BT-DO3A) and gadoteridol (with HP-DO3A).
- Group C: The chelating agent is pyclen, including gadopidlenol.

Information regarding potential CR among GBCA is very limited and data on the recently introduced gadopidlenol not yet available.

Similar to ICM, we propose an empirical selection of an alternative GBCA according to the chemical structure of the chelating agent. However, we cannot include gadopidlenol as an alternative until data about CR are available, although it seems promising due to its different chelating agent.

In the case of known GBCA:

- When the implicated CM is from Group A: Prioritize the use of one from Group B.
- When the implicated CM is from Group B: Prioritize the use of one from Group A.

In the case of an unknown GBCA:

GBCA from Group B should be prioritized due to both the ability of gadoterate meglumine to induce immediate non-allergic HSR through an immunoglobulin-independent mechanism, including Mas-Related G-Protein-coupled Receptor X2 (MRGPRX2) activation [42], and its ionic nature.

Outside the European Union, when a macrocyclic GBCA is not available, a linear GBCA could be used. However, in the case of a reaction with type A GBCA (gadoterate meglumine), the use of ionic linear GBCA

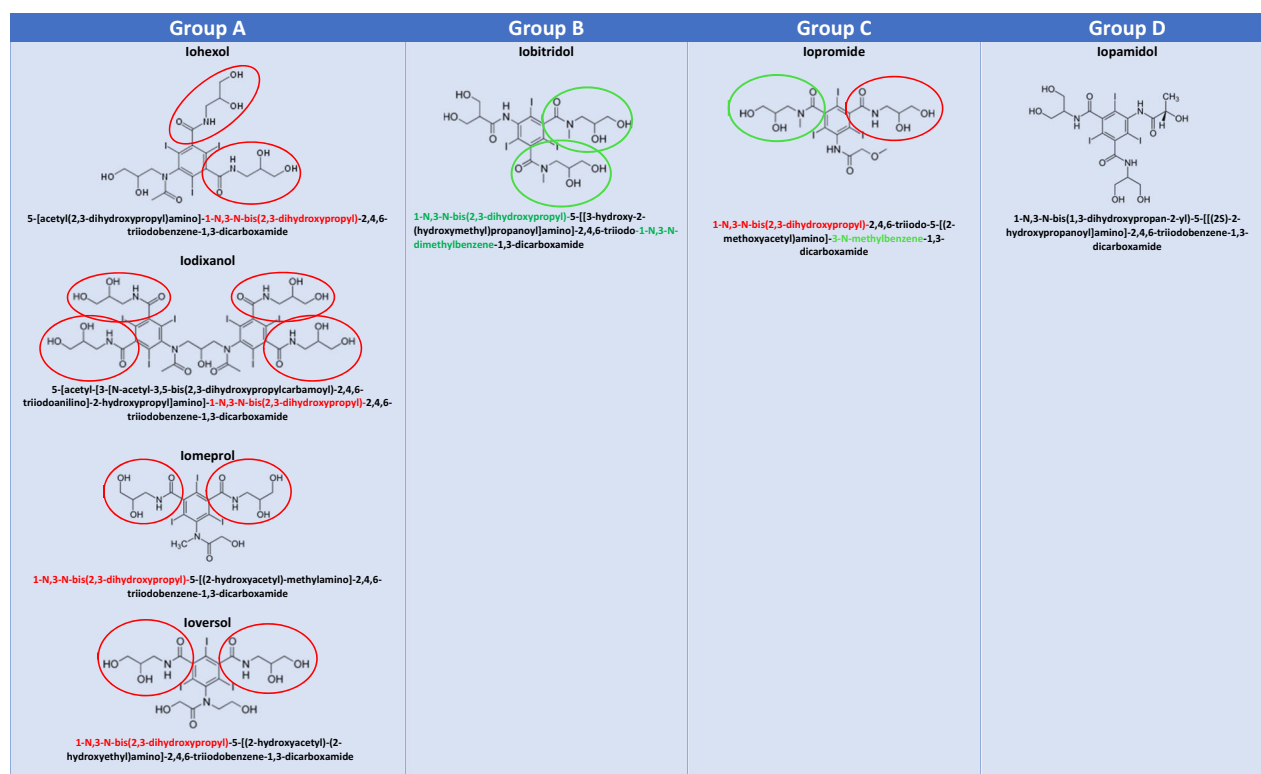


Fig. 1 Chemical formulas and comparison of molecule and benzene chain arrangements by colours. Position 1 and 3 of the benzene ring: Classic carbamoyl sidechain (1-N,3-N-bis(2,3-dihydroxypropyl) in red. Modified carbamoyl sidechain in green. Position 2, 4, 6 of the benzene ring: three iodine atoms. Position 5 of the benzene ring: highly variable side chains defining diverse types of ICM

with a meglumine-cation (gadopentetate and gadobenate) should be avoided. This recommendation is based on the fact that meglumine can activate mast cells through a mechanism not mediated by MRGPRX2 [42].

In the event of a positive allergy study with all tested CMs or when the administration of a CM with questionable tolerance is urgent, a desensitization protocol may be considered. However, there is very limited data regarding the protocol that should be utilized and its safety [43, 44]. Finally, an empirical selection of a CM entails a certain risk of eliciting new reactions, thus necessitating an individualized assessment of the benefit of a new CM administration. Both the radiologist and the prescribing physician should be involved in the decision-making process. The patient, or their family members, if necessary, should be informed about the risks of using CM and may even need to complete an informed consent form.

In conclusion, we present a novel approach for empirically selecting an alternative CM in the case of previous HSR. Although an allergy evaluation is preferred, in situations where this cannot be done, a proposal based on chemical structures of the different CM could help.

Regardless, when the alternative CM has been chosen empirically, heightened vigilance and preparedness to address potential recurrent reactions is necessary because the possibility of inducing an HSR is higher than if the selection has been guided by an allergy study.

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Methodology

- Retrospective literature review

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